Create Database and Tables:

```sql
CREATE DATABASE IF NOT EXISTS hospital_db;
USE hospital_db;

CREATE TABLE Patients (
    patient_id INT PRIMARY KEY,
    patient_name VARCHAR(100) NOT NULL,
    age INT,
    gender VARCHAR(10),
    blood_group VARCHAR(5),
    city VARCHAR(50),
    registration_date DATE
);

CREATE TABLE Doctors (
    doctor_id INT PRIMARY KEY,
    doctor_name VARCHAR(100) NOT NULL,
    specialization VARCHAR(50),
    consultation_fee DECIMAL(10,2)
);

CREATE TABLE Appointments (
    appointment_id INT PRIMARY KEY,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    appointment_date DATE,
    status VARCHAR(20),
    FOREIGN KEY (patient_id)
        REFERENCES Patients(patient_id),
    FOREIGN KEY (doctor_id)
        REFERENCES Doctors(doctor_id)
);

CREATE TABLE Prescriptions (
    prescription_id INT PRIMARY KEY,
    appointment_id INT NOT NULL,
    medication VARCHAR(100),
    dosage VARCHAR(50),
    duration_days INT,
    FOREIGN KEY (appointment_id)
        REFERENCES Appointments(appointment_id)
);
```

Inserted Values into Tables

patient_id	patient_name	age	gender	blood_group	city	registration_date
1	Aarav Sharma	28	Male	O+	Chennai	01/10/2025
2	Diya Iyer	34	Female	B-	Bengaluru	02/15/2025
3	Rohan Mehta	45	Male	A+	Mumbai	03/20/2025
4	Ananya Rao	12	Female	AB+	Chennai	04/05/2025
5	Karthik Nair	52	Male	O-	Kochi	05/12/2025
6	Meera Krishnan	39	Female	B+	Hyderabad	06/18/2025
7	Arjun Patel	23	Male	A-	Mumbai	07/22/2025
8	Sneha Das	31	Female	O+	Kolkata	08/09/2025
9	Vikram Singh	61	Male	AB-	Delhi	09/14/2025
10	Priya Menon	26	Female	O-	Bengaluru	10/25/2025

doctor_id	doctor_name	specialization	consultation_fee
201	Dr. Kavitha Raman	Cardiology	1500.00
202	Dr. Rahul Verma	Dermatology	900.00
203	Dr. Suresh Kumar	Orthopedics	1200.00
204	Dr. Nisha Thomas	Pediatrics	800.00
205	Dr. Arvind Rao	Neurology	1800.00
206	Dr. Priya Shah	ENT	700.00
207	Dr. Manoj Das	General Medicine	500.00
208	Dr. Sneha Kapoor	Ophthalmology	1000.00
209	Dr. Deepak Nair	Dentistry	600.00
210	Dr. Anjali Menon	Physiotherapy	750.00

appointment_id	patient_id	doctor_id	appointment_date	status
5001	1	201	03/02/2026	Completed
5002	2	202	03/03/2026	Completed
5003	3	203	03/05/2026	Completed
5004	1	204	03/08/2026	Completed
5005	4	205	03/10/2026	Completed
5006	5	201	03/12/2026	Completed
5007	6	206	03/14/2026	Cancelled
5008	7	207	03/15/2026	Completed
5009	8	202	03/18/2026	Completed
5010	2	208	03/20/2026	Completed

prescription_id	appointment_id	medication	dosage	duration_days
9001	5001	Atorvastatin	10 mg daily	30
9002	5002	Cetirizine	10 mg daily	5
9003	5003	Ibuprofen	200 mg as needed	3
9004	5004	Paracetamol	500 mg as needed	3
9005	5005	Levetiracetam	250 mg twice daily	30
9006	5006	Aspirin	75 mg daily	30
9007	5008	Amoxicillin	500 mg three times daily	5
9008	5009	Loratadine	10 mg daily	7
9009	5010	Lubricating Eye Drops	As directed	7

SELECT with WHERE and ORDER BY

```
5   USE hospital_db;
6
7   SELECT doctor_name, specialization, consultation_fee
8   FROM Doctors
9   WHERE consultation_fee > 900
10  ORDER BY consultation_fee DESC;
```

Output | Result #2

doctor_name	specialization	consultation_fee
Dr. Arvind Rao	Neurology	1800.00
Dr. Kavitha Raman	Cardiology	1500.00
Dr. Suresh Kumar	Orthopedics	1200.00
Dr. Sneha Kapoor	Ophthalmology	1000.00

Multiple conditions using AND, OR, and IN

```sql
SELECT patient_id, patient_name, city, blood_group
FROM Patients
WHERE city IN ('Chennai', 'Bengaluru')
  AND blood_group IN ('O+', 'O-')
ORDER BY patient_name;
```

Output | Result #1

patient_id	patient_name	city	blood_group
1	Aarav Sharma	Chennai	O+
10	Priya Menon	Bengaluru	O-

INNER JOIN

```sql
SELECT
    a.appointment_id,
    p.patient_name,
    d.doctor_name,
    a.appointment_date,
    a.status
FROM Appointments a
INNER JOIN Patients p
    ON a.patient_id = p.patient_id
INNER JOIN Doctors d
    ON a.doctor_id = d.doctor_id
ORDER BY a.appointment_date;
```

Output | Result #1

appointment_id	patient_name	doctor_name	appointment_date	status
5001	Aarav Sharma	Dr. Kavitha Raman	03/02/2026	Completed
5002	Diya Iyer	Dr. Rahul Verma	03/03/2026	Completed
5003	Rohan Mehta	Dr. Suresh Kumar	03/05/2026	Completed
5004	Aarav Sharma	Dr. Nisha Thomas	03/08/2026	Completed
5005	Ananya Rao	Dr. Arvind Rao	03/10/2026	Completed
5006	Karthik Nair	Dr. Kavitha Raman	03/12/2026	Completed
5007	Meera Krishnan	Dr. Priya Shah	03/14/2026	Cancelled
5008	Arjun Patel	Dr. Manoj Das	03/15/2026	Completed
5009	Sneha Das	Dr. Rahul Verma	03/18/2026	Completed
5010	Diya Iyer	Dr. Sneha Kapoor	03/20/2026	Completed

Joining all four tables

```sql
SELECT
    a.appointment_id,
    p.patient_name,
    d.doctor_name,
    a.appointment_date,
    pr.medication,
    pr.dosage,
    pr.duration_days
FROM Appointments a
INNER JOIN Patients p
    ON a.patient_id = p.patient_id
INNER JOIN Doctors d
    ON a.doctor_id = d.doctor_id
INNER JOIN Prescriptions pr
    ON a.appointment_id = pr.appointment_id
ORDER BY a.appointment_date;
```

Output | Result #1

appointment_id	patient_name	doctor_name	appointment_date	medication	dosage	duration_days
5001	Aarav Sharma	Dr. Kavitha Raman	03/02/2026	Atorvastatin	10 mg daily	30
5002	Diya Iyer	Dr. Rahul Verma	03/03/2026	Cetirizine	10 mg daily	5
5003	Rohan Mehta	Dr. Suresh Kumar	03/05/2026	Ibuprofen	200 mg as needed	3
5004	Aarav Sharma	Dr. Nisha Thomas	03/08/2026	Paracetamol	500 mg as needed	3
5005	Ananya Rao	Dr. Arvind Rao	03/10/2026	Levetiracetam	250 mg twice daily	30
5006	Karthik Nair	Dr. Kavitha Raman	03/12/2026	Aspirin	75 mg daily	30
5008	Arjun Patel	Dr. Manoj Das	03/15/2026	Amoxicillin	500 mg three times daily	5
5009	Sneha Das	Dr. Rahul Verma	03/18/2026	Loratadine	10 mg daily	7
5010	Diya Iyer	Dr. Sneha Kapoor	03/20/2026	Lubricating Eye Drops	As directed	7

Aggregation using GROUP BY and HAVING

```sql
SELECT
    d.doctor_name,
    d.specialization,
    COUNT(a.appointment_id) AS completed_appointments,
    SUM(d.consultation_fee) AS total_consultation_fees
FROM Doctors d
INNER JOIN Appointments a
    ON d.doctor_id = a.doctor_id
WHERE a.status = 'Completed'
GROUP BY d.doctor_id, d.doctor_name, d.specialization
HAVING SUM(d.consultation_fee) > 1500
ORDER BY total_consultation_fees DESC;
```

Output | Result #1

doctor_name	specialization	completed_appointments	total_consultation_fees
Dr. Kavitha Raman	Cardiology	2	3000.00
Dr. Rahul Verma	Dermatology	2	1800.00
Dr. Arvind Rao	Neurology	1	1800.00

Nested query using AVG()

```sql
SELECT doctor_name, specialization, consultation_fee
FROM Doctors
WHERE consultation_fee > (
    SELECT AVG(consultation_fee)
    FROM Doctors
)
ORDER BY consultation_fee DESC;
```

Output | Result #1

doctor_name	specialization	consultation_fee
Dr. Arvind Rao	Neurology	1800.00
Dr. Kavitha Raman	Cardiology	1500.00
Dr. Suresh Kumar	Orthopedics	1200.00
Dr. Sneha Kapoor	Ophthalmology	1000.00

Nested query with a correlated subquery

```sql
SELECT
    p.patient_id,
    p.patient_name,
    (
        SELECT SUM(d.consultation_fee)
        FROM Appointments a
        INNER JOIN Doctors d
            ON a.doctor_id = d.doctor_id
        WHERE a.patient_id = p.patient_id
    ) AS total_consultation_fees
FROM Patients p
WHERE p.patient_id IN (
    SELECT a.patient_id
    FROM Appointments a
    INNER JOIN Doctors d
        ON a.doctor_id = d.doctor_id
    GROUP BY a.patient_id
    HAVING SUM(d.consultation_fee) > 1500
)
ORDER BY total_consultation_fees DESC;
```

Output | Result #1

patient_id	patient_name	total_consultation_fees
1	Aarav Sharma	2300.00
2	Diya Iyer	1900.00
4	Ananya Rao	1800.00